

Predicting High-Usage NBA Players (Season 2015 - 2024)

Xiaoyi Lu

Data Science Initiative, Brown University

[Github](#)

1. Introduction

Purpose

In NBA competition, usage rate (USG%) serves as an important indicator of a player's offensive involvement. Players with high usage rates typically play roles on the main scoring and offensive responsibilities for the team, while players with low usage rates usually play more supportive, off-ball roles. Therefore, accurately identifying which players have high usage rates is crucial for team tactical analysis and player management. Modeling and analyzing whether a player fits the high-usage role based on their individual statistics at the season level is an effective and highly insightful research method.

NBA Player-Seasonal Statistics Dataset (Kaggle & Basketball-Reference)

The NBA player-season data used in this project comes from Kaggle and Basketball Reference, covering season-level statistics for NBA players from the 2015 to 2024 seasons. This dataset contains basic and advanced statistical metrics, including points, field goal attempts, assists, and rebounds, to comprehensively characterize a player's offensive involvement in games. The data is organized on a "player-season" basis, with each row representing a player's overall performance in a single season. [1]

Target Variable and Features

The NBA player-season data used in this project is a binary classification dataset. The research objective is to determine whether a player is a high-usage player in a given season. The target variable is defined as 'is_high_usg'. When a player's usage rate is greater than or equal to 0.25, the variable is marked as 1, indicating a high-usage player. Otherwise, it is marked as 0. The dataset contains approximately 5,500 player-season samples and over 30 feature variables used for modeling. The feature variables mainly consist of basic and advanced statistical metrics at the player-season level, including playing time, field goal attempts, rebounds, various efficiency, and usage-related indicators. These features are all numerical variables used to characterize a player's overall offensive involvement. In addition, the data also includes categorical features such as season information and player identity to distinguish performance differences across different seasons and individual players.

Current Research

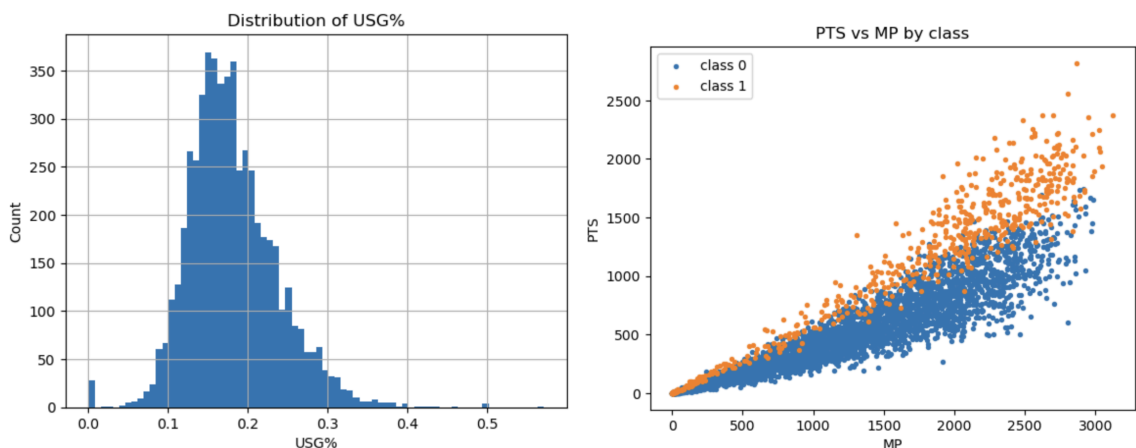
In recent years, NBA data analytics research has increasingly used Usage Percentage (USG%) as an important indicator to measure a player's offensive involvement, characterizing their role in the offensive system [2]. Existing research indicates that USG% more accurately reflects a player's actual participation in offensive possessions than simply considering playing time, but

its impact varies across different game situations and time periods [3]. Some studies have found that high-usage players may dominate the offense early in a game, but later tend to experience decreased efficiency or increased turnovers due to fatigue [2]. On the other hand, regarding the relationship between usage rate and long-term player efficiency improvement, empirical analysis has shown that a high USG% in a rookie season does not consistently correlate with significant efficiency improvements in subsequent seasons [3]. Therefore, existing research still lacks a systematic modeling and analysis of the formation mechanisms of high-usage NBA players from the perspectives of multi-season data, imbalanced classification, and model interpretability.

2. Exploratory Data Analysis

Feature Analysis

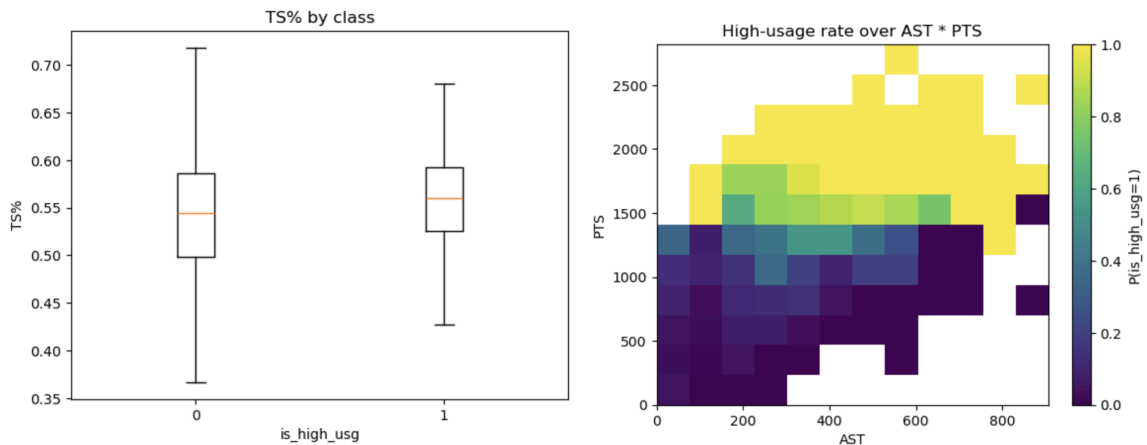
I used descriptive statistics to examine the data dimensions, feature ranges, and missing values of the dataset. Some basic statistics (such as points and rebounds) showed a moderate amount of missing values, while advanced metrics showed higher missing rates, indicating that missing values were widespread across various features. I plotted a histogram of Usage Percentage (USG%) to understand the distribution of the target variable. The main peak was between 0.10 and 0.22, with only a few players exceeding 0.30. Based on this distribution, I used 0.25 as a threshold to define the binary target variable 'high_usg'.



The scatter plot of Points (PTS) against Minutes Played (MP) showed a strong positive correlation between scoring output and playing time. High-usage players generally scored more points than non-high-usage players at similar playing times, suggesting they took on greater offensive responsibility.

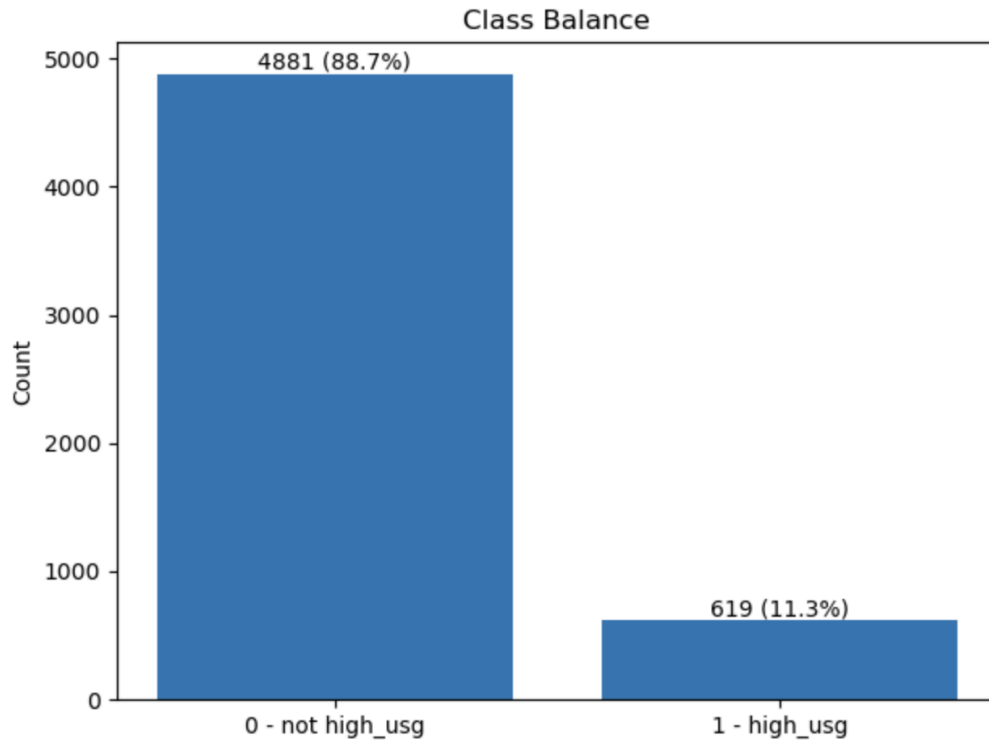
The box plot comparison of True Shooting Percentage (TS%) showed that high-usage players had slightly higher median efficiency, but the distribution ranges largely overlapped. This suggests that higher usage rate does not

necessarily indicate to higher efficiency. The heatmap of Assists and Points indicated that players with both high scoring and playmaking abilities were more likely to be classified as high-usage players.



Target Variable

The target variable in this dataset is 'is_high_usage'. Each value is defined whether each player's usage rate (USG%) exceeds a predefined threshold (i.e., usage rate greater than or equal to 25%). Each value is classified as either '0' or '1', representing non-high usage and high usage, respectively. Exploratory analysis shows that high-usage players constitute a small proportion of the sample, indicating it is an imbalance data. This distribution characteristic means that relying solely on accuracy may not fully reflect model performance. Therefore, subsequent modeling and evaluation will focus on evaluation metrics more suitable for imbalanced classification problems, such as the F1-score.



3. Method

[Data Preprocessing]

Dataset Split

In this study, the dataset was divided according to the chronological order of the seasons to preserve the temporal structure of NBA player performance data. Player season data from 2015 to 2020 was used as the training set, data from the 2021-2022 was used as the validation set, and data from 2023 to 2024 was reserved as the test set, resulting in an approximate split ratio of 60:20:20. This time-series splitting strategy helps prevent information leakage and better reflects real-world prediction scenarios.

To address the issue of imbalanced distribution of the target variable, the class distribution was kept consistent across the training, validation, and test sets. Model hyperparameters were selected based on the performance on the validation set, and the final evaluation was performed on the reserved test set.

Preprocessing

This dataset contains both numerical and categorical features, and missing values are present in several statistical measures. For numerical features, missing values are imputed using the median. For categorical features, missing values are imputed using the most frequent category. This method preserves all observations without discarding samples or features.

All numerical features are standardized using a standard scaler to bring them to a comparable scale. Standardization does not restrict the numerical range and therefore does not distort the original distribution of the data. Categorical features, including season and player identifiers, are transformed using one-hot encoding.

Features

After constructing the target variable, the original USG% feature was removed from the feature input set to avoid data leakage.

The final feature set includes player personal information, playing time, scoring statistics, efficiency metrics, and offensive playmaking-related metrics, as well as categorical season and player features. After preprocessing and encoding, the transformed feature space will be used for all models.

ML Pipeline

Four machine learning algorithms are used in this analysis: Logistic Regression, Random Forest, Gradient Boosting, and SVM classifier algorithm. For each algorithm, several parameters were tuned. (Table 1)

Logistic Regression	C : [0.001, 0.01, 0.1, 1, 10, 100, 1000]
Random Forest	n_estimators: [100, 200] max_depth: [3, 5, 10, None] max_features: [0.3, 0.7, 1.0]
Gradient Boosting	learning_rate: [0.05, 0.1] max_depth: [2, 3] n_estimators: [100, 200]
SVM	C: [0.1, 1, 10, 100, 1000] gamma: [0.001, 0.01, 0.1, 1, 10]

<Table 1. Tuned parameter values for each machine learning algorithm>

The primary metric used to evaluate model performance in this study is the F1 score. Due to the imbalanced distribution of the target variable `is\_high\_usg`, accuracy alone is not a reliable measure of model performance. The F1 score provides a balanced evaluation, considering both precision and recall, making it more suitable for this classification task.

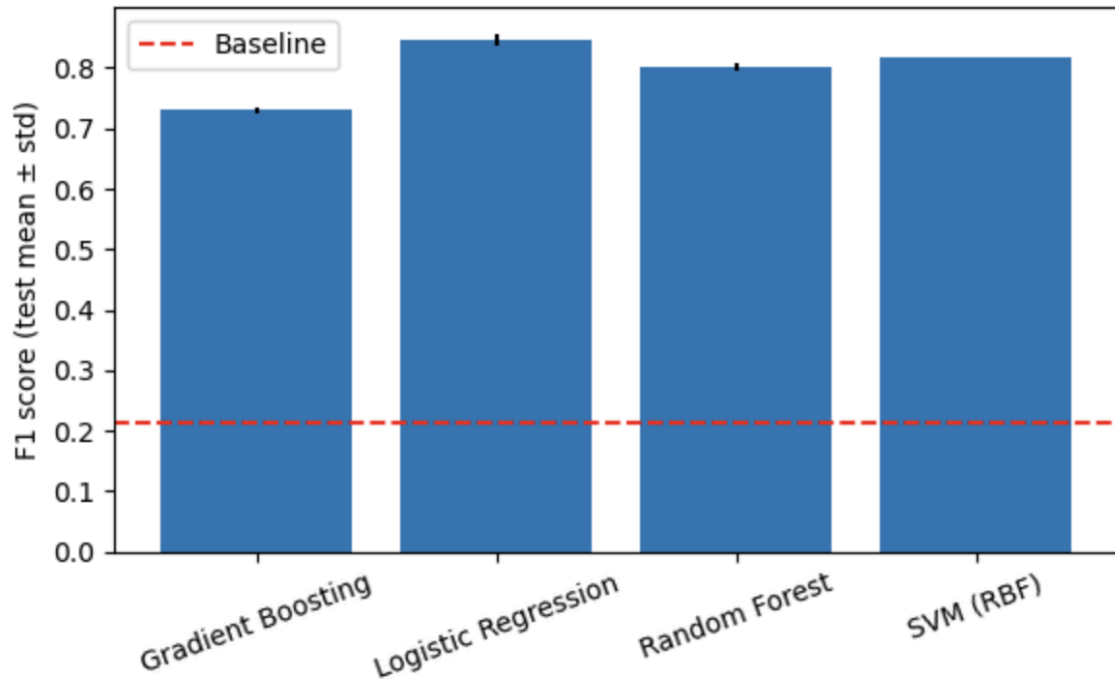
In this analysis, each machine learning model was evaluated under five different random states. For each random state, model hyperparameters were selected based on validation set performance, and the corresponding test performance was recorded. Most models showed little variation in test F1 scores across different random states, indicating stable results and insensitivity to random initialization. Deterministic models such as logistic regression and support vector machines produced identical results across different runs, while tree-based models exhibited limited variability.

Compared to the baseline with minority class, applying cross-validation model selection improved the performance of Logistic Regression, Random Forest, and SVM models. The baseline model, which ignored feature information, obtained significantly lower F1 scores, indicating that the learned models were able to capture meaningful patterns in the data. All reported test results are from the held-out test set to ensure a fair comparison between models.

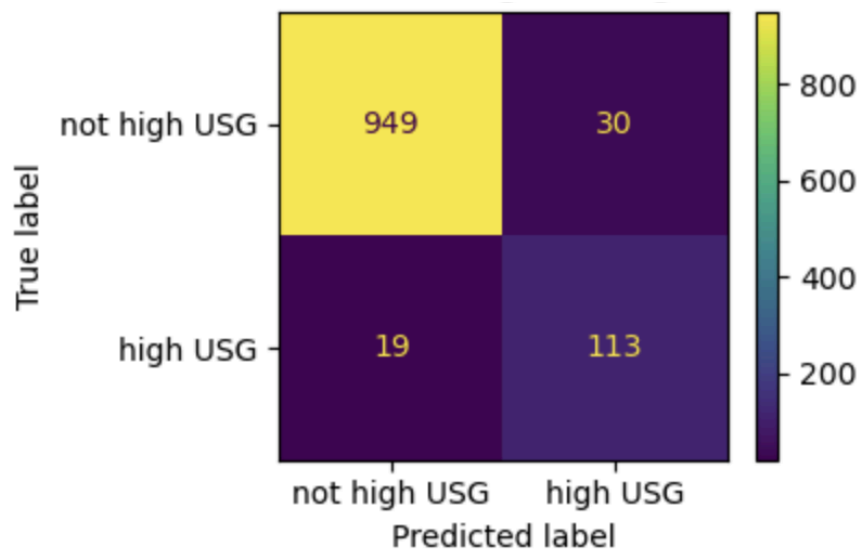
4. Results

	Accuracy		F1_Score		Recall	
	Mean	Std	Mean	Std	Mean	Std
Model						
Baseline (Minority Class)	0.119	NaN	0.212	NaN	1.000	NaN
Logistic Regression	0.956	0.000	0.822	0.000	0.856	0.000
Random Forest	0.950	0.008	0.787	0.019	0.779	0.031
Gradient Boosting	0.944	0.002	0.722	0.015	0.612	0.024
SVM	0.958	0.000	0.817	0.000	0.795	0.000

<Table 2: Performance Comparison Across Models>



<Table 3: Model Performance>



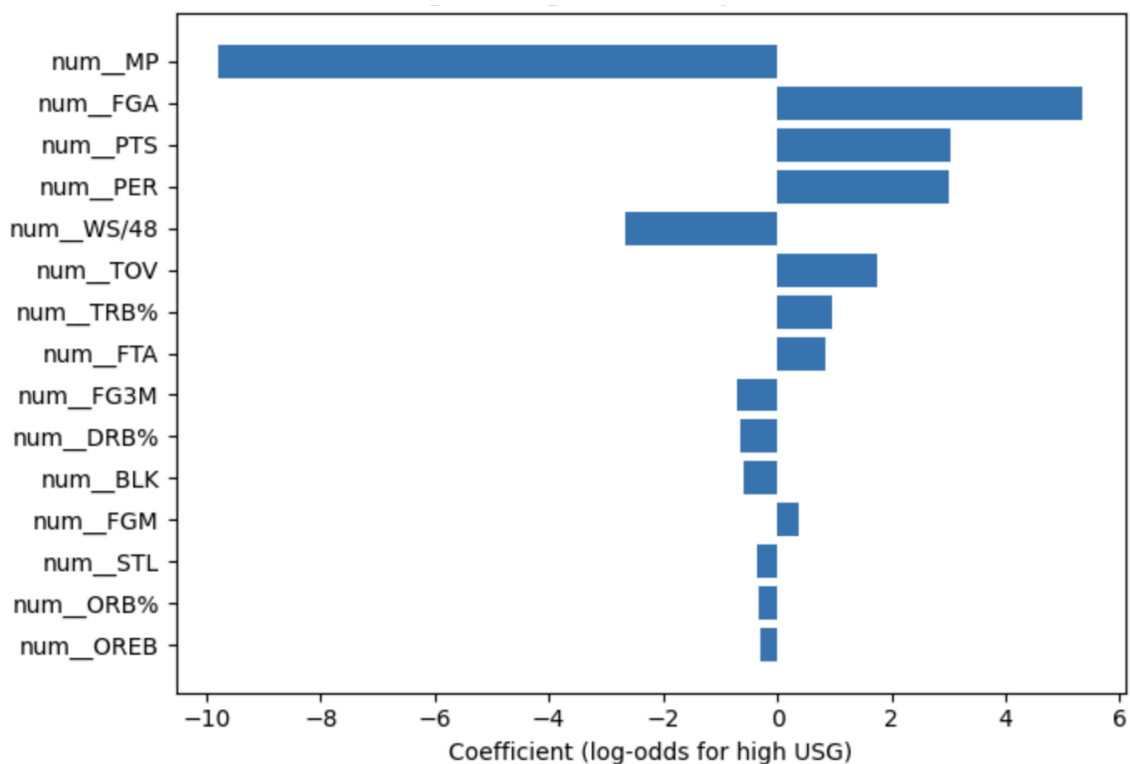
<Table 4: Test set confusion matrix using the best performing Logistic Regression model>

The baseline within the minority-class achieved an F1 score of 0.212, establishing a significant class imbalance problem in the dataset. All machine learning models performed significantly better than the baseline model, indicating

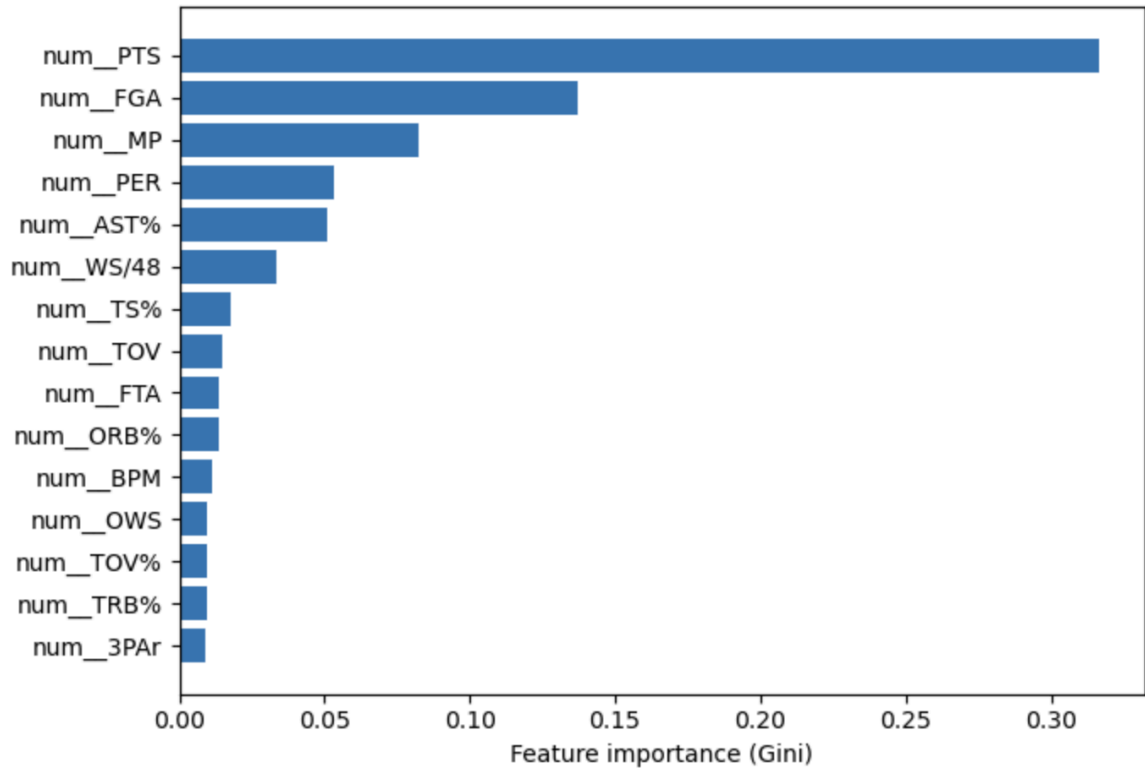
that the input features provided meaningful information for distinguishing between high-usage and non-high-usage players. Among all models, Logistic Regression achieved the highest average F1 score on the test set, while the Gradient Boosting model had the lowest overall performance.

Figure 3 compares the F1 scores of different models on the test set. Logistic Regression consistently outperformed other methods, followed closely by SVM and Random Forest model. The small standard deviations across runs indicate that the models performed stably under different random states.

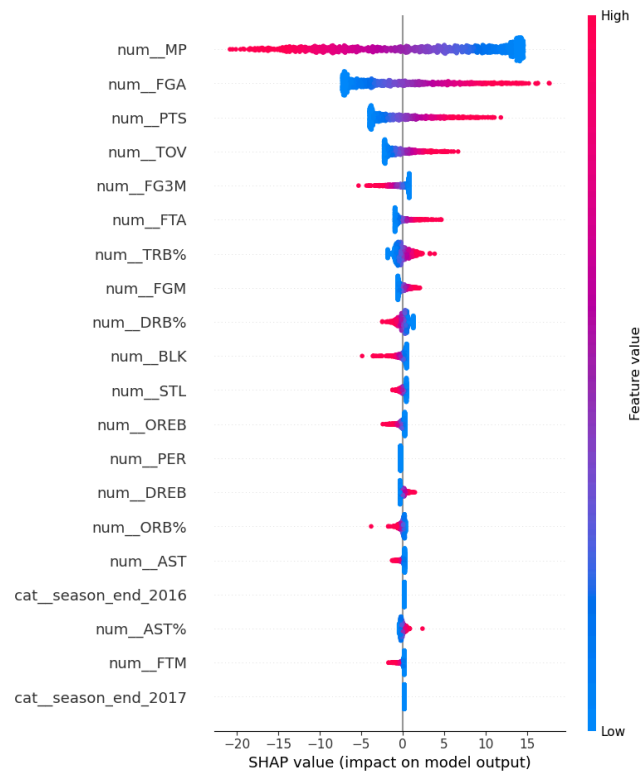
Figure 4 shows the confusion matrix for the best-performing Logistic Regression model on the test set. The model correctly classified most non-high-usage players and identified most high-usage players, with relatively few misclassifications. This result demonstrates that the model achieved a good balance between precision and recall for the minority class.



<Table 5 : Top 15 Coefficient Importances within Logistic Regression>



<Table 6 : Top 15 Important features in Random Forest within Gini Importance>



<Table 7 : Impact of Top 20 features in Logistic Regression prediction>

Coefficient values from Logistic Regression, Gini importance from Random Forest, and SHAP values are used to interpret which features contribute most to predicting high-usage players. Across these three approaches, scoring and usage-related features consistently appear among the most influential predictors.

According to the Logistic Regression coefficient analysis (Table 5), minutes played (MP), field goal attempts (FGA), points scored (PTS), and usage-related efficiency metrics exhibit the largest absolute coefficients, indicating strong associations with high-usage roles. Features with positive coefficients increase the likelihood of a player being classified as a high-usage player, while negative coefficients correspond to lower usage probability.

The Random Forest model based on Gini importance (Table 6) similarly highlights points (PTS), field goal attempts (FGA), and minutes played (MP) as the most important features. Although the ranking differs slightly from the Logistic Regression coefficients, both models emphasize scoring volume and offensive involvement as key determinants of high usage.

The SHAP analysis for the Logistic Regression model (Table 7) provides additional insight into feature effects at the individual prediction level. Higher values of MP, FGA, and PTS generally contribute positively to the prediction of high usage, while lower values reduce the predicted probability. The consistency between SHAP results and coefficient-based interpretations further supports the robustness of these findings.

5. Outlook

In this study, Logistic Regression performed best among all models, exhibiting high predictive performance, stability across different random states, and good interpretability. This model achieved ideal results with only limited tuning of the regularization parameter. Future research could explore different regularization forms or other linear classification models to improve flexibility while maintaining interpretability.

Feature analysis results indicate that high-usage players are primarily characterized by features related to offensive involvement, such as playing time, field goal attempts, and points scored. This conclusion is consistent across coefficient analysis, random forest feature importance, and SHAP explanations.

Future research could incorporate more contextual variables (e.g., team pace, lineup structure, or player position) to more comprehensively characterize the mechanisms underlying high-usage roles.

Although this study used multi-season player-level data, further incorporating game-by-game or play-by-play data, or analyzing the dynamic changes in player usage over time, could provide a more detailed understanding of the evolution of high-usage roles in different contexts.

6. Reference

- [1] *NBA Top players + Deep Learning*. Piyush. Retrieved from October 25, 2025, from <https://www.kaggle.com/code/piyush1912/nba-top-players-deep-learning/notebook#Random-Forest-+-GridSearchCV>
- [2] Iatropoulos, D., Sarlis, V., & Tjortjis, C. (2025, March 25). A Data Mining Approach to Identify NBA Player Quarter-by-Quarter Performance Patterns. <https://www.mdpi.com/2504-2289/9/4/74>
- [3] Tabibian, R. (2022, April 7). Do Higher Minutes Played and Higher Usage Percentage in a Player's Rookie Year Result in Improved Efficiency in Their Second and Third Years? <https://sites.dartmouth.edu/sportsanalytics/2022/04/07/do-higher-minutes-played-and-higher-usage-percentage-in-a-players-rookie-year-result-in-improved-efficiency-in-their-second-and-third-years/>